

Listing 11-7: The `Balloon` classes

The `Balloon` class is an abstract class ❶, so the `BalloonMgr` instantiates objects (randomly) from the `BalloonSmall` ❷, `BalloonMedium` ❸, and `BalloonLarge` ❹ classes. Each of those classes creates a `pygame` `Image` object, then calls the `__init__()` method in the `Balloon` base class. We differentiate the balloons with the arguments representing the image, size, number of points, and speed.

The `__init__()` method in the `Balloon` class ❺ stores the information about each balloon in instance variables. We get the rectangle of the balloon image and remember its width and height. We set a randomized horizontal position that will ensure that the balloon image will fully show within the window.

Every time a `MOUSEDOWNEVENT` happens, the balloon manager loops through the `Balloon` objects and calls the `clickedInside()` method of each ❻. The code here checks to see if the `MOUSEDOWNEVENT` that was detected happened inside the current balloon. If it did, the `Balloon` plays the pop sound and returns a Boolean to say that it was clicked on, as well as the number of points that balloon was worth. If it was not hit, it returns `False` and zero.

In each frame, the balloon manager calls the `update()` method of each `Balloon` ❼, which updates that `Balloon`'s `y` position by subtracting its own speed in order to move higher in the window. After changing the position, the `update()` method returns either `BALLOON_MISSED` (if it has moved completely off the top of the window) or `BALLOON_MOVING` (to indicate that it is still in play).

The `draw()` method simply draws the image of the balloon at the appropriate `(x, y)` location ❽. Although the `y` position is kept as a floating-point value, `pygame` automatically converts it to an integer for pixel placement in the window.

The last method, `__del__()` ❹, has been added for debugging and for future development. Whenever the balloon manager deletes a balloon, the `__del__()` method of that `Balloon` object is called. For demonstration purposes, for now it simply prints a message that displays the balloon's size and ID number.